Customer Shopping Behavior Analysis

End-to-end data analytics report — Python/Pandas cleaning, MySQL querying, and Power BI dashboarding on 3,900 customer transactions.

Prepared by	Hassan Jumaa (Riq)
Role	Data Analyst
Dataset	Customer Shopping Behavior (3,900 transactions, 18 fields)
Tools used	Python (Pandas), MySQL, Power BI
Repository	github.com/Riq-wq/Customer-Shopping-Behavior-Analysis

1. Executive Summary

This report analyzes 3,900 customer transactions to evaluate purchasing patterns, customer segments, and subscription behavior. The data was cleaned using Python (Pandas), loaded into MySQL, and queried to extract ten core business insights. Average spend is $59.76 across 3,900 unique customers.

Total Transactions	Unique Customers	Avg Spend	Male Revenue	Female Revenue
3,900	3,900	$59.76	$157,890.00	$75,191.00

Key Takeaways

- Male customers generate $157,890.00 in revenue — more than double female customers at $75,191.00.
- Subscribers average $59.49 per purchase vs $59.87 for non-subscribers — subscription does not drive higher spend.
- Loyal customers (3,116 or 79.9%) dominate the base — retention is the key growth lever.
- Hat has the highest discount rate at 50.0% — discount-dependent products need margin review.
- Express shipping customers average $60.48 vs $58.46 for Standard — express users spend more per order.

2. Methodology & Data Pipeline

The analysis followed a three-stage pipeline: cleaning → SQL analysis → visualization.

- **Cleaning (Python/Pandas):** Loaded raw CSV (3,900 rows, 18 columns). Filled 37 nulls in `Review Rating` with category median. Standardized column names to snake_case. Engineered `age_group` and `purchase_frequency_days`. Dropped `promo_code_used` (perfectly correlated with `discount_applied`). Saved cleaned CSV with 17 columns, 0 nulls, 0 duplicates.
- **Loading (MySQL):** Cleaned DataFrame loaded into MySQL table `customer_shopping_behavior` via SQLAlchemy + PyMySQL.
- **Querying (SQL):** Ten targeted business questions answered directly in MySQL (see `sql/customer-shopping-behavior.sql`).
- **Visualization (Power BI):** Interactive dashboard built on the cleaned dataset.

3. Revenue by Gender

Male customers generate $157,890.00 (67.7% of total) vs female customers at $75,191.00 (32.3%). This 2.1:1 revenue ratio suggests product mix or purchasing frequency differs significantly by gender — worth investigating for targeted marketing.

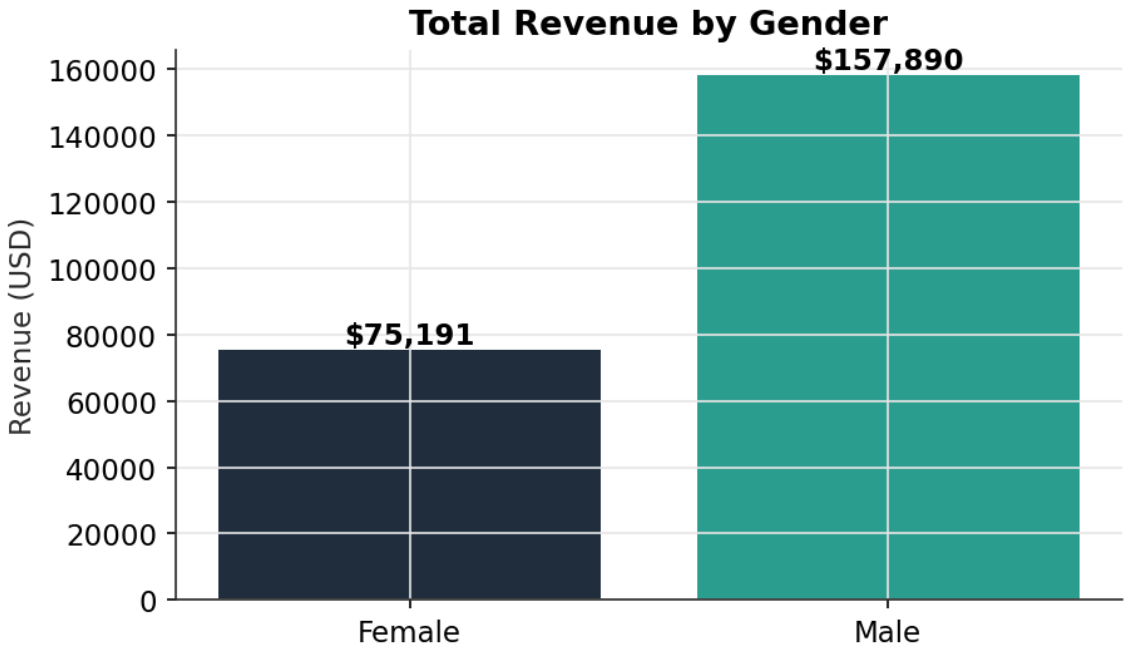


Figure 1. Total Revenue by Gender

4. Subscription Behavior

Subscribers (1,053 customers) average $59.49 per purchase, while non-subscribers (2,847 customers) average $59.87. The difference is minimal (0.6%), indicating subscription status alone is not a strong predictor of spend. However, subscribers contribute $62,645.00 vs $170,436.00 for non-subscribers.

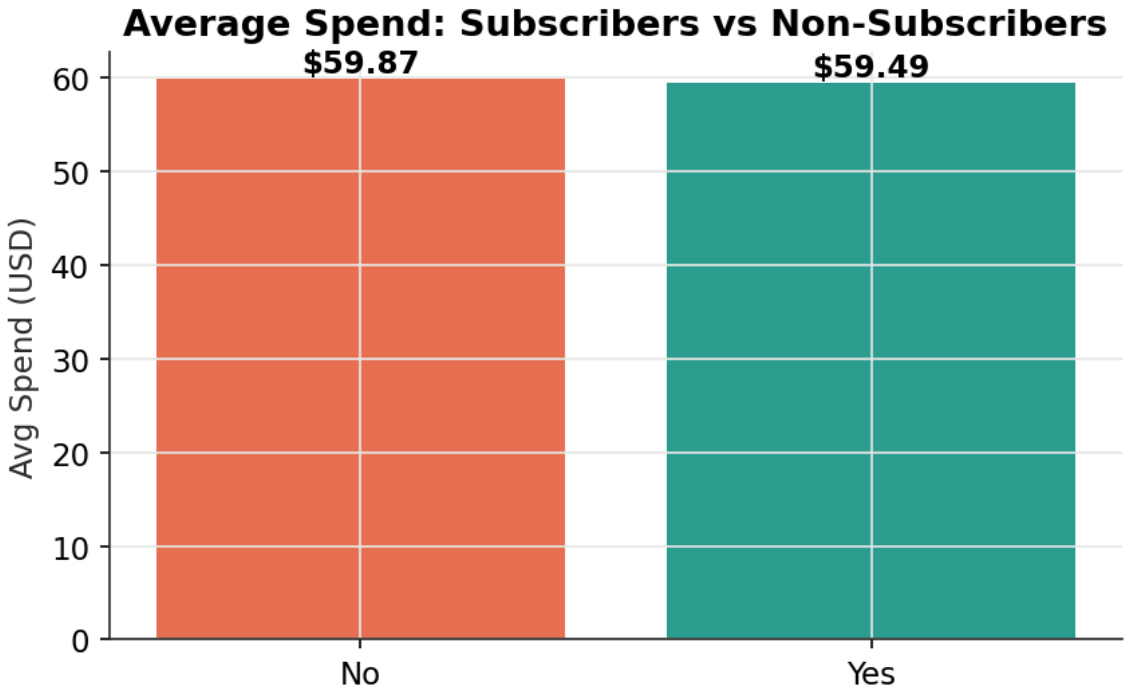


Figure 2. Average Spend: Subscribers vs Non-Subscribers

5. Customer Segmentation

Customers segmented by previous purchases: **Loyal** (3,116 customers, 79.9%), **Returning** (701 or 18.0%), **New** (83 or 2.1%). With 79.9% of customers in the Loyal segment, retention programs are more valuable than acquisition.

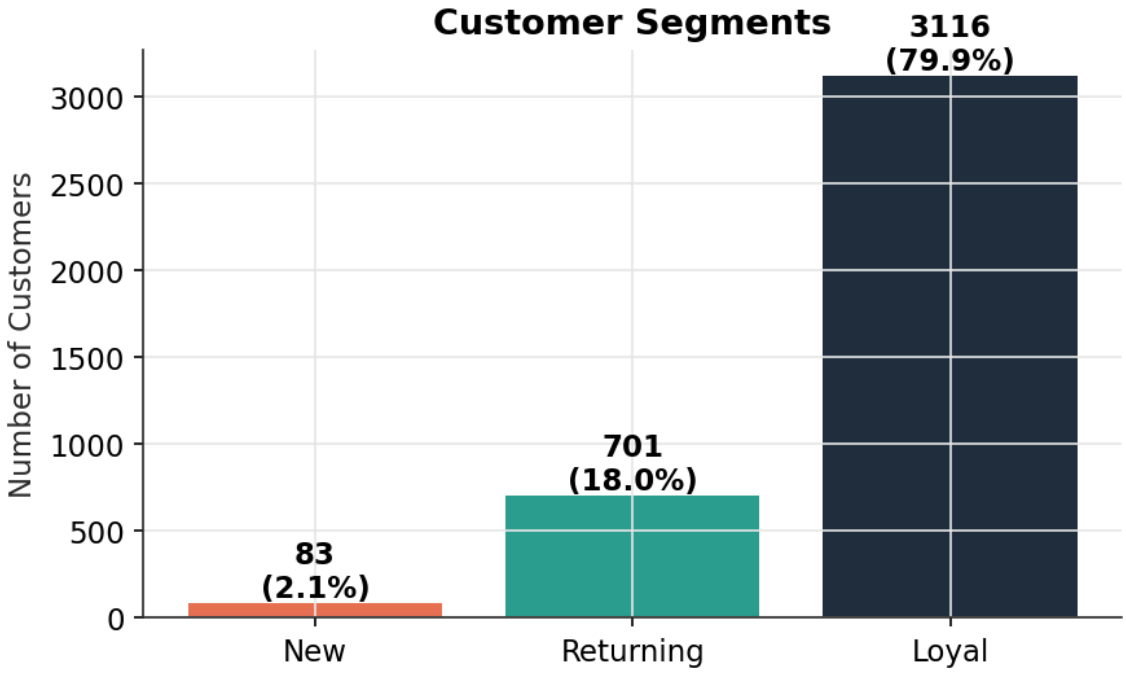


Figure 3. Customer Segments

6. Discount Analysis

Hat has the highest discount rate at 50.0%. Discount-dependent products erode margin and should be reviewed for pricing strategy. 839 customers used discounts while spending above the average purchase amount of $59.76.

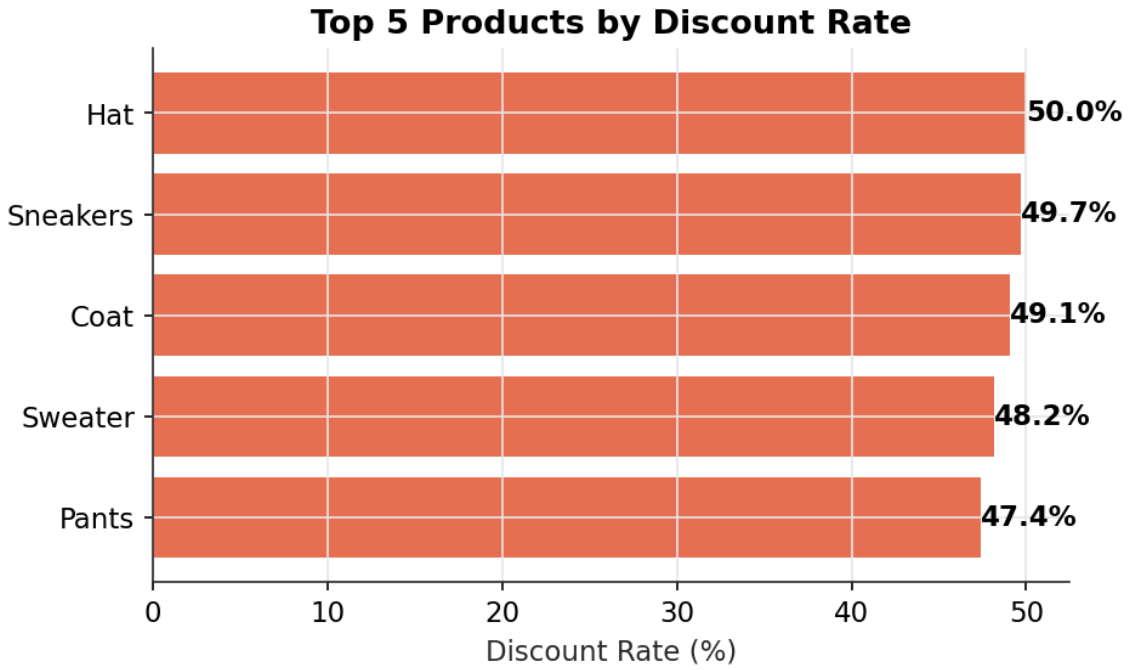


Figure 4. Top 5 Products by Discount Rate

7. Shipping Behavior

Express shipping customers average $60.48 per order vs $58.46 for Standard — a 3.4% premium. Express users are willing to pay more for faster delivery, suggesting a price-insensitive segment for premium service tiers.

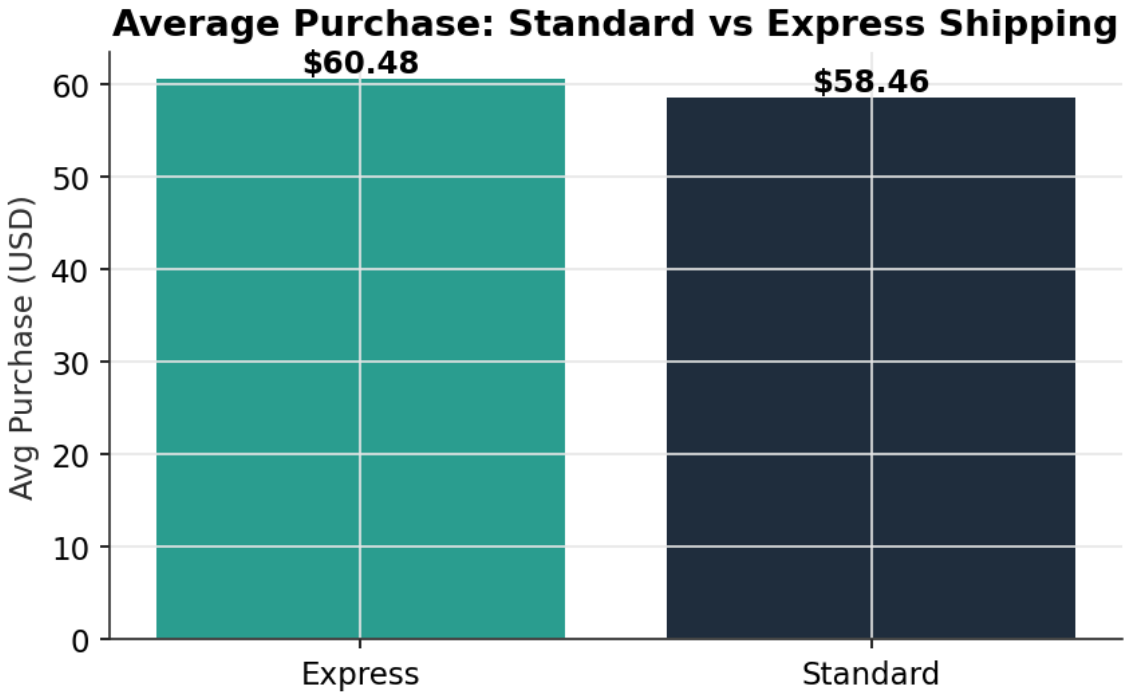


Figure 5. Average Purchase: Standard vs Express Shipping

8. Revenue by Age Group

Young Adults generates the highest revenue at $62,143.00. Marketing and inventory should align with the spending power of each age cohort.

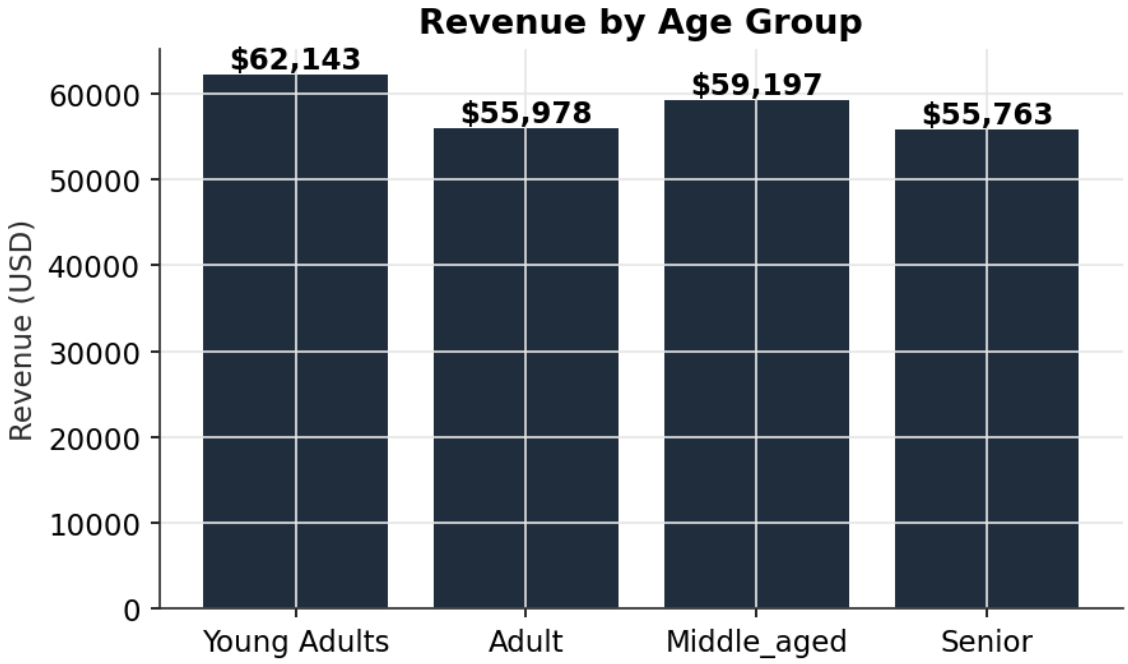


Figure 6. Revenue by Age Group

9. Recommendations

- **Boost subscriptions.** Subscribers don't spend more per order, so focus on exclusive benefits that increase purchase frequency rather than discounting.
- **Loyalty programs.** 79.9% of customers are already Loyal — reward repeat buyers to protect this core segment.
- **Review discount policy.** Discount-dependent products (e.g., Hat at 50.0% discount rate) need margin controls.
- **Target express users.** Express shipping customers spend 3.4% more per order — premium service tiers are worth pursuing.
- **Targeted marketing.** Focus on high-revenue age groups and gender segments identified in this analysis.

10. Conclusion

Customer behavior analysis reveals that retention drives revenue more than acquisition. With 79.9% of customers in the Loyal segment and minimal spend difference between subscribers and non-subscribers, the business should focus on loyalty programs, targeted marketing for express users, and disciplined discounting to protect margins. The accompanying Power BI dashboard makes these patterns explorable in real time.